

Problem 1

Let $N_t \sim \text{Poisson}(t)$ be the Poisson process of weekly plague arrivals and let $p = 0.9$ represent the probability that the Pharaoh reneges on his promise.

- Since Pharaoh's reneging on his promise is independent from the plague arrival process, by thinning we can split the process into 2 independent Poisson processes $N_t^f \sim \text{Poisson}(pt)$ and $N_t^s \sim \text{Poisson}((1-p)t)$, tracking the arrivals of plagues that failed and succeeded to convince the Pharaoh to let the Israelites go. Thus, the time it takes before the Israelites finally leave corresponds to the first arrival of the Poisson process of plagues that succeeded in convincing the Pharaoh: $T_1^s \sim \text{Exp}(1-p) = \text{Exp}(0.1)$.
- We can characterize the event in question as an intersection of two independent events: $A = \{\text{the first 10 arrivals of the plague process are all of different types}\}$, and $B = \{\text{Israelites leave on the 10th plague}\}$. Since each type is equally likely to be chosen for each plague arrival, we can compute the probability of A by counting the number of favorable outcomes $\mathbb{P}(A) = \frac{10!}{10^{10}}$, since there are $10!$ many ways permute 10 different numbers in 10 slots. As for event B , let $B_i = \{i\text{-th plague arrival succeed to convince the Pharaoh}\}$, thus we get:

$$\mathbb{P}(B) = \mathbb{P}(\{\cap_{i=1}^9 B_i^C\} \cap B_{10}) = \mathbb{P}(B_{10}) \prod_{i=1}^9 \mathbb{P}(B_i^C) = (1-p)p^9 = \frac{9^9}{10^{10}}.$$

Thus, for the event in question we get:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) = \frac{9^9 \cdot 10!}{10^{20}} \approx 1.4 \cdot 10^{-5}.$$

Grading Comments:

- part a: Most students can identify the Poisson process N_t of plague arrivals with the correct rate. It is important to use the thinning property to get the Poisson process for Israelites to leave. Many students give similar arguments, but some failed to do that.

The common mistake comes from the distribution of the time it takes before the Israelites finally leave Egypt. Lots of students didn't connect this event to the first arrival time of a Poisson distribution.

- part b: Students struggled on this part, trying to write some words or properties, which leads to nowhere. Most student didn't identify the two independent events in the solution. To calculate each probability, the common mistake is that they didn't consider the permutation of the plagues.

Problem 2

Let $L_r \sim \text{Exp}(\lambda_r)$ and $L_s \sim \text{Exp}(\lambda_s)$ be the exponential random variables tracking the length of the rising and proofing stages respectively, where $\lambda_r = \frac{1}{8}$ and $\lambda_s = \frac{1}{4}$.

- a. Since L_r and L_s are independent exponential random variables we know that their minimum is also exponential with the additive rate, that is $L_r \wedge L_s \sim \text{Expon}(\lambda_r + \lambda_s)$. Since $L_r \vee L_s = L_r + L_s - L_r \wedge L_s$ we get

$$\mathbb{E}[L_r \vee L_s] = \mathbb{E}[L_r] + \mathbb{E}[L_s] - \mathbb{E}[L_r \wedge L_s] = \frac{1}{\lambda_r} + \frac{1}{\lambda_s} - \frac{1}{\lambda_r + \lambda_s} \approx 9.33.$$

- b. Using the law of total probability and the fact that L_r and L_s are independent we get:

$$\begin{aligned} \mathbb{P}(L_r + L_s \leq 12) &= \int_0^\infty \mathbb{P}(L_r + L_s \leq 12 | L_s = t) \mathbb{P}(L_s \in dt) \\ &= \int_0^\infty \mathbb{P}(L_r + t \leq 12) \mathbb{P}(L_s \in dt) = \int_0^\infty \left(1 - e^{-\lambda_r(12-t)}\right) 1_{\{0 \leq t \leq 12\}} \lambda_s e^{-\lambda_s t} dt \\ &= 1 - e^{-12\lambda_s} - \frac{\lambda_s e^{-12\lambda_r}}{\lambda_s - \lambda_r} (1 - e^{-12(\lambda_s - \lambda_r)}) = 1 - e^{-3} - 2e^{-1.5}(1 - e^{-1.5}) \\ &= (1 - e^{-1.5})^2 \approx 0.6. \end{aligned}$$

Grading Comments:

- Part a: Many students think $\text{Exponential}(\lambda)$ has a mean λ , however, the mean should be $\frac{1}{\lambda}$.
- Part b: In this part we aim to solve for the probability of $L_r + L_s \leq 12$. Although L_r and L_s are independent, the events $L_r = a$ (where a is a constant where $0 < a < 12$) and $L_r + L_s \leq 12$ are not independent since once L_r is set to be a , then L_s must be less than $12 - a$ to satisfy the requirement that $L_r + L_s \leq 12$. Some students either think distribution of the sum of two independent exponentially distributed variables should be used, or that $L_r = a$ and $L_r + L_s \leq 12$ are independent.

Problem 3

Let $S_n = S_0 + \sum_{i=1}^n X_i$ be the lazy random walk tracking the movements of the Israelites on the interval $[0, 4]$, where $S_0 = 1$ and for all i :

$$X_i = \begin{cases} 1 & \text{w.p. } \frac{2}{30} := p, \\ 0 & \text{w.p. } \frac{27}{30} := s, \\ -1 & \text{w.p. } \frac{1}{30} := q. \end{cases}$$

- a. This event A is equivalent to Israelites making 3 steps to the right and staying once at the same spot. There are 3 choices for the time we need to stall, therefore the event A is equal to the disjoint union of events A_i which correspond to stalling at the i -th step, where $i \in [1, 3]$, and going to the right on the remaining 3 steps. In particular,

$$A_i = \{X_i = 0 \text{ and } X_j = 1, j \neq i\},$$

and since X_i 's are iid we get:

$$\mathbb{P}(A_i) = \mathbb{P}(A_1) = \mathbb{P}(X_1 = 0) \prod_{i=2}^4 \mathbb{P}(X_i = 1) = \frac{1}{3750} \approx 2.67 \cdot 10^{-4}.$$

Therefore, $\mathbb{P}(A) = 3\mathbb{P}(A_1) = \frac{1}{1250} = 8 \cdot 10^{-4}$.

- b. Let T denote the time Israelites spend wandering in the wilderness. We can use first step analysis to calculate $\mathbb{E}[T|S_0 = 1]$, for that denote $\mathbb{E}[T|S_0 = i] = r(i)$ and observe that:

$$\begin{aligned} r(i) &= 1 + pr(i+1) + sr(i) + qr(i-1) \Rightarrow \\ r(i) &= \frac{1}{p+q} + \frac{p}{p+q}r(i+1) + \frac{q}{p+q}r(i-1). \end{aligned}$$

Note that $r(4) = r(0) = 0$, thus the above system of equations reduces to three linear equations:

$$\begin{cases} r(1) &= 10 + \frac{2}{3}r(2), \\ r(2) &= 10 + \frac{1}{3}r(1) + \frac{2}{3}r(3), \\ r(3) &= 10 + \frac{1}{3}r(2). \end{cases} \Leftrightarrow \begin{cases} r(2) &= 20 + \frac{4}{9}r(2), \\ r(1) &= 10 + \frac{2}{3}r(2), \\ r(3) &= 10 + \frac{1}{3}r(2). \end{cases}$$

Solving the first equation we get $r(2) = 36$ and therefore $r(1) = 10 + \frac{2}{3}36 = 34$. Thus it will take on average the Israelites 34 years of travel in the wilderness until they reach either Egypt or Israel.

Grading Comments:

- Part a: Most student can successfully identify there are 3 different choices, while some students think there are 4 choices rather than 3.
- Part b: Most students can find the right direction: either using the one-step analysis to solve a series of equations or using the general formula for the bias random walk, while most students cannot solve the equations correctly or use the wrong formula.